MILITARY
SPECIFICATION

MIL-M-13787(SigC)
Amendment No. 1
17 November 1955

MOTORS, ALTERNATING CURRENT, FRACTIONAL HORSEPOWER, SQUIRREL CAGE (FOR COMMUNICATION AND OTHER ELECTRONIC AND SPECIAL MILITARY APPLICATIONS)

1. Paragraph 2.1 Specifications.

Delete:

72-53 Finishes (for Ground Signal Equipment)

Add:

MIL-L-3545 Lubricating Grease; High Temperature
MIL-L-14072 Finishes for Ground Signal Equipment
MIL-L-15719 Lubricating Grease (High Temperature, Electric Motor, Ball and Roller Bearings).

2. Paragraph 3.2.5 Lubricants. Delete entire paragraph and substitute:

"Lubricants shall conform to Specifications MIL-G-3278, MIL-L-3545, MIL-L-15719, or MIL-L-6085."

3. Paragraph 3.3.1.1 Bearing mounting. Change sentence to read as follows:

"Unless otherwise specified, when bearings are mounted in soft metal such as zinc alloys, aluminum or copper, a steel insert shall be provided."

4. Add the following paragraph:

"3.3.1.2 Heat stabilization. Bearings to be operated above 90°C shall be heat stabilized so that their dimensions do not increase more than 0.0001 inch per inch in diameter during 2500 hours at 149° F."

5. Add the following paragraph:

"3.3.1.3 Sleeve bearings. Sleeve bearings shall have an oil reservoir adequate to permit 1000 hours of operation between lubrication periods."

6. Paragraph 3.3.2 Impregnation. Delete the third sentence and replace with the following:

"For class A and B insulation, impregnation varnish shall conform to Specification MIL-V-1137, Grade CB, unless otherwise specified by the contracting officer. For class H insulation, silicone impregnating varnish shall be used."

FSC 6105

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7. Add the following paragraph:

"3.3.3 Rotor. Motors shall be so constructed that the rotor can be readily removed without the use of special tools."

8. Add the following paragraph:

"3.9.2.3 Tapped holes. Where tapped holes are used for mounting or assembly, suitable protection shall be included in the design of the motor to prevent damage to the motor winding should an extra long screw be used."

9. Paragraph 3.9.6.1 Continuous. Add as last sentence, "Unless otherwise specified, the motor shall meet the requirements for full-load speed and torque after the 2500 hours life test specified in paragraph 4.2.16."

10. Paragraph 3.9.6.2 Intermittent. Add as last sentence, "Unless otherwise specified, the motor shall meet the requirements for full-load speed and torque after the 1000 hours life test specified in paragraph 4.2.16."

11. Paragraph 3.21 Moisture resistance. Delete, "48 hour drying period", and substitute, "24 hour running period."

12. Paragraph 3.24 Marking. Delete paragraph and substitute, "The motor shall be Permanently and legibly marked with the designation "MOTOR, ALTERNATING CURRENT" the Government drawing of specification sheet number, the manufacturer's name or symbol, horsepower, duty cycle, full load speed, voltage, full load current, direction of rotation, serial no., capacitor rating (if applicable) and wiring diagram. Terminal marking shall conform to Specification JAN-E-251. The direction of rotation shall be plainly and permanently indicated by an arrow for unidirectional and reversible motors."

13. Paragraph 3.26 Delete item 10, "Life".

14. Paragraph 3.27 Serial numbers. Delete entire paragraph.

15. Paragraph 4.1.2 Standard test conditions. Add after "herein", or on the complimentary documents."

16. Paragraph 4.2.3 Insulation resistance. Change to read as follows:

"Measurements shall be made at 500 volts d.c. The test voltage shall be applied between ground (frame, positive instrument lead) and each winding (negative instrument lead) which is not permanently grounded by soldering, brazing, or welding, with all other windings grounded. Windings which are normally connected together in service are considered as one winding for the purpose of this test. A second reading using reversed instrument polarity (negative instrument lead grounded and positive instrument lead to the winding)

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shall be made. The lowest reading shall be considered the insulation resistance. (See 3.11)."

17. Add the following paragraph:

"4.2.5.1 Air-gap measurements. For motors having a rating above 1/4 horse-power, the uniformity of the Air-gap shall be determined by ascertaining that the rotor turns freely in the assembled motor, when wound with a wire spaced spirally around the rotor periphery. For this test, the diameter of the wire shall be 70 percent of the nominal air-gap of the motor."

18. Paragraph 4.2.7.3 Efficiency. Delete first sentence.

19. Paragraph 4.2.7.4 No load. Delete first sentence.

20. Paragraph 4.2.8 Temperature rise. After "Temperature rise". Delete the sentence following.

21. Paragraph 4.2.10 Vibration. Under "(c) Motion" Change, "7-1/2 hours" to "6 hours"; "2-1/2 hours" to "2 hours"; and "1-1/4 hours" to "1 hour."

22. Paragraph 4.2.11.1.2 Number of blows. Last sentence, change height of hammer blows from "2 feet, 3 feet, and 4 feet" to "1 foot, 3 feet, and 5 feet."

23. Paragraph 4.2.13 Overload. Delete first sentence.

24. Paragraph 4.2.15(d) Moisture resistance. Third sentence, change "1 hour at full-load torque", to "24 hours at full-load torque", after "full-load speed", add "and insulation resistance." Delete 4th sentence.

25. Paragraph 4.2.16, 4.2.16.1 and 4.2.16.2 Delete and substitute:

"4.2.16 Life. During life test, precautions shall be taken against undue air contamination of a type not to be expected in normal service use. The motors shall be operated at rated voltage and output torque at its specified duty cycle. Continuous duty motors shall be operated for 2500 hours and intermittent duty motors shall be operated for 1000 hours. Unless otherwise specified on the complimentary document, the life test shall be run at an ambient temperature of 65°C (150°F). Undirectional continuous duty motors shall be stopped and started every 24 hours. Reversible continuous duty motors shall be reversed every 24 hours. Reversible intermittent duty motors shall be alternately operated in each direction at rated duty cycle. Following this test, the motor shall pass the full load speed and torque test at room ambient temperature. No maintenance of any kind shall be allowed during this test except that sleeve bearings may be lubricated at not less than 1000 hours intervals."

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26. Paragraph 6.9 Frame sizes. Change paragraph to read as follows:

"6.9 NEMA frame sizes. In order to accomplish standardization of motors for military applications, motors should conform to the NEMA standard frame sizes indicated in Figure 3 wherever possible."

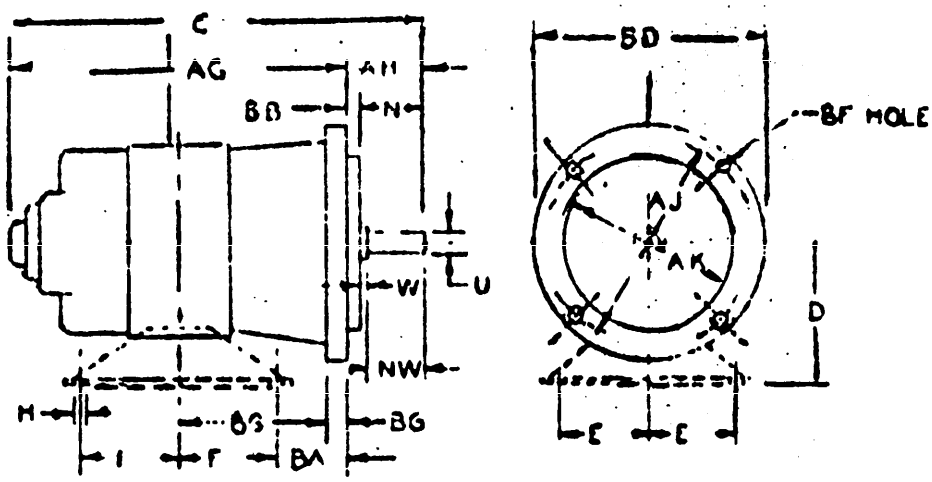
27. Figure 1 Motor-Identification Form. Add as Item 26 the following:

"26 specialized test conditions (if applicable)"

28. Figure 3 Change subtitle from "(From NEMA Standard MG1-3.2)" to "(From NEMA Standard MG1-3.02)".

29. Add Figure 3a.

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Frame No.	U	AH	AJ	AK	BB	BC	BC Max.	BG
42C	3/8	1-5/16	3-3/4	3	5/32 Max.	3/16	5	2-23/32
84C	1/2	1-11/16	3-3/4	3	5/32 "	3/16	5-5/8	3-11/16
56C	5/8	2-1/16	5-7/8	4-1/2	5/32 "	3/16	6-1/2	4-1/16
66C	3/4	2-7/16	5-7/8	4-1/2	5/32 "	3/16	6-1/2	5-7/16

BF Hole

Frame No.	Number	Tap Size
42C	4	1/4-20
48C	4	1/4-20
56C	4	3/8-16
66C	4	3/8-16

All Dimensions in inches

TOLERANCES

AK Dimension - 42-66 frames, incl. - +0.000, -0.003 inch.
Face Runout - 42-66 frames, incl. - -0.004 inch indicator reading.
Permissible Eccentricity of Mounting Rabbet - 42-66 frames, incl. - -0.004 inch indicator reading.

Notes:
N-W is the length of shaft extension from shoulder to shaft end.

Standard Dimension for Type C Face-mounted, Direct and Alternating-current Motors

(From NEMA Standard MG1-3.06)

Figure 3a

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Tolerances specified for group A, group B, and group C (see tables V, VI and VII,

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